UNIGUARD DIGITAL WALLET

AI-Driven Personal Budgeting and Financial Literacy Enhancement for University Students

"Turning Pocket Money into Smart Money"

INTRODUCTION

UniGuard Wallet is a user-friendly web application created specifically for university students in Uganda who often face money challenges. Most rely on unpredictable income from allowances or small gigs, but without proper budgeting skills or financial know-how, they end up overspending on things like airtime and outings. As noted by the National Council for Higher Education (2018), this frequently causes debt, anxiety, and lower grades. UniGuard makes it simple with instant spending tracking, smart AI that builds realistic budgets for irregular income, enjoyable gamification that teaches money management, and an AI companion for personalized advice. It helps students take control, spend smarter, feel less stressed, develop strong habits, and stay focused on their studies so they can finish university confident and financially secure.

ABSTRACT

Financial illiteracy and poor budgeting habits among youth and students remain persistent global challenges. This research project focuses on helping university students and young people manage their money better. Many students depend on small or irregular sources of income, such as pocket money, allowances, gifts, or part-time jobs. Because of this, they often struggle to plan their spending, save money, or avoid overspending. Many also lack financial knowledge, which makes the problem worse. UniGuard Wallet addresses these challenges through an AI-powered personal finance system with smart budgeting, receipt scanning, predictive analytics, and gamified financial literacy.

PROBLEM STATEMENT

University students in Uganda face severe financial challenges due to low financial literacy (only 34–38% of Ugandans are financially literate – S&P Global FinLit Survey 2015; InterMedia FII), poor budgeting habits, and dependence on irregular income from allowances and casual gigs. This leads to impulse spending, mounting debt, high stress levels, and academic underperformance (National Council for Higher Education, 2018). The lack of accessible, student-focused financial tools creates a critical gap, preventing young people from building responsible money habits during their university years.

PROPOSED SOLUTION

- **Smart Budgeting System:** Track income types, categorize expenses, set spending limits via Budget Ports
- **Budget Optimization Engine:** Distinguish essential vs. non-essential expenses, suggest weekly budgets using AI allocation
- **AI Receipt Scanner:** DocTR OCR extracts merchant, total, and line items from receipt photos
- **Financial Literacy Gamification:** Achievements, rewards, leaderboards
- **Predictive Analytics:** Spending forecast, goal prediction, anomaly detection (logic-based)
- **Time Machine:** Simulate future financial states from current trends

OBJECTIVES

- Develop a smart budgeting tool tailored to students (Web Application)
- Provide AI-driven budget optimisation and overspending prediction
- Deliver interactive financial literacy gamification
- Build a secure and lightweight system with AI receipt scanning

PRELIMINARY RESULTS

Implemented: Full web app with dashboard (Wallet Hub), transactions, budget ports, goals, AI companion, reports, achievements, leaderboard. DocTR-based receipt scanner; logic-based anomaly detection and predictive forecasting; Time Machine simulator; bill splitting, savings vault, family finance, investment tracking modules.

ALIGNMENT WITH SDGS & NDP IV

SDG 1 – No Poverty: Build lasting financial stability.

SDG 4 – Quality Education: Financial literacy gamification.

SDG 8 – Decent Work & Economic Growth: Smart money habits.

National Development Plan (NDP IV): Aligns with the Digital Economy Program and Human Capital Development.

CONCLUSION

UniGuard Wallet offers a practical and empowering solution to the financial struggles faced by university students, especially those relying on unpredictable income. By combining smart budgeting tools, real-time tracking, AI receipt scanning, predictive analytics, and engaging financial learning features, it equips users with the skills and discipline needed to manage money confidently. Ultimately, UniGuard helps students build lasting financial stability, reduce stress, and stay focused on achieving academic success.

FUTURE WORKS

- Integration with mobile money APIs for automatic transaction syncing
- Deploying reinforcement learning for adaptive budgeting
- Advanced behavioural analytics dashboards
- Beta testing with UCU students
- Expansion to full cross-university deployment
- ML-based transaction categorizers and per-user anomaly model (UGX)

REFERENCES

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UNIGUARD DASHBOARD

Wallet Hub with quick stats, financial wellness score, recent transactions, budget preview, smart predictions feed, anomaly summary, spending alerts.

Dashboard preview: Quick Stats | Financial Nexus | Budget Ports | Goals

SYSTEM ARCHITECTURE

Frontend
React, Vite, Tailwind, Shadcn

→

Backend-AI
Flask, receipt parsing

→

DocTR API
OCR / KIE

→

Supabase
Auth, DB, storage

Local logic-based models for real-time predictions; ML pipeline for training categorizers.